

Occurrence and Triggers of Mobile News Exposure: A Screenshot-Based ESM Study

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ABSTRACT

Mobile phone has a variety of apps and services where users can encounter news. However, existing research mainly focused on news exposure on specific apps and has relied on retrospective methods. Thus, the connection between news exposure and their triggers remains unclear. We conducted a screenshot-based ESM study and have recruited twelve users who ran the study for fourteen days. Our results show that users' different news exposures were associated with different kinds of triggers. For example, news exposure on content recommender systems was mainly triggered by the need for exploring news, whereas news exposure on messaging apps was mainly triggered by notifications. Future research can continue to study how context influences triggers and news exposure.

CCS CONCEPTS

• Human-centered computing → Mobile phones.

KEYWORDS

mobile news, news consumption, motivation, multiplatform, screenshots, experience sampling

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1 INTRODUCTION

With the advent of pervasive mobile internet, smartphones have reportedly become a primary medium people access news in many

countries, such as U.S. [7], German [9], as well as our country, Taiwan [6]. The increasing exposure to news on smartphones has drawn many researchers' attention to study how such exposure differs from on other media [3]. Nevertheless, despite a variety of apps and services on which smartphones users can encounter news, such as social media, news apps, browsers, search engines, video apps, or the content recommender offered by the smartphone system (e.g. Google Discover¹), the existing body of research either has been focused on users' news exposure on only specific apps (e.g. social media [4], browser [1]), or relies on retrospective methods to recall news exposure [5], which is subject to recall bias. Due to the limitation of these methods, there has been a lack of empirical evidence describing: 1) the *in situ* occurrences of news exposure via different apps, and 2) the triggers of these news exposures; that is, what brings users to see the news on these different apps, and how the trigger(s) to these news exposures differ if any?

To answer these two questions, we aim to connect *in situ* news exposure with their triggers to yield a better understanding of mobile news consumption. We leverage automated screenshots [2] to capture the dynamics of smartphone users' news exposure across apps, accompanied with an experience sampling method (ESM) that asks users about the triggers of specific sampled news exposure and the context under which it takes place. This screenshot-based ESM approach allows us to obtain not only in-the-moment news exposure data but also their associated triggers and contexts.

We have recruited twelve smartphone users who participated in our study for fourteen days. Due to the page limit, in this paper, we mainly report on our preliminary observations of the connection between these users' news exposures and their triggers. At a high level, we show that different kinds of news exposure were associated with different kinds of triggers. For example, news exposure on news/content recommender system (e.g. Google Discover) was more often triggered by users' habitual use of the phone or the motivation to know more about recent news; news exposure on social media apps was more often triggered by the initial motivation for relaxing, and news exposure on messaging apps are more often triggered by phone notifications.

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¹<https://blog.google/products/search/introducing-google-discover/?fbclid=IwAR0pTyL5Oqg1n4j4PVDYafv3OiyTh7vk6jfp1gHqp713JNa2sKAtZhxno>

2 METHOD

2.1 Research Instrument

We developed an Android research app that automatically detected the use of targeted apps, including *social media apps* (Facebook and Instagram), *messaging apps* (Messenger and Line), *content Recommender apps* (Google Discover and LineToday), *Browsers*, *online video apps* (Youtube), and *News Apps*. When users open one of the apps above except *messaging apps*, the research app requests users' consent to take screenshots that take place every second. For *messaging apps*, the research app detects and records URLs contained in the messages, without taking any screenshots. To ask users to reflect on news exposure, the app monitors users' uses of targeted apps and randomly chooses one of the most recent apps uses for prompting users to answer an ESM questionnaire. Users receive 3-11 ESM prompts per day, and the interval between any two questionnaires is at least one hour long. An ESM prompt is dismissed if it is not responded to within 15 minutes.

In each ESM questionnaire, users are provided screenshots (URLs for messaging apps) that are sampled since the last ESM they respond to and choose the most recent screenshot that contains the news they have an impression on. The purpose is to leverage screenshots as a cue for helping users recall and reconstruct the moments when they encounter news, such that their responses in the ESM would be more accurate and reliable. After choosing a screenshot, users are asked to answer the primary trigger(s) of the selected news exposure, including: *To obtain more news information*, *Out of habit*, *For entertainment*, *For relaxing or killing time*, *Notified by the phone*, *For study or work purpose* and *For social* [8].

2.2 Recruitment and Participants

Currently, we have recruited twelve participants via social media posts (gender: 8 = female, 4 = male; age: $M = 21.67$, $SD = 1.3$; occupation: 11 = college students, 1 = employed). Every participant attended a physical pre-study meeting, where researchers introduced the study and helped them install the research app. In two weeks study period, we instructed them to use their phone and consume news as they would normally do in their daily lives and to fill out the ESM when they were prompted to. Upon completion of the study, we invited our participants to a post-study interview to understand their news consumption experiences.

3 FINDINGS

A total of 855 ESM responses were received from the participants. 386 replied "I didn't encounter news in this period" in the first question. Among the remaining 469 ESM questionnaires ($M=39.1$, $SD=13.2$, $Min=24$, $Max=68$), participants accessed news on *social media* (50.75%), *content recommender* (32.62%), *messaging App* (13.43%), *browser* (1.92%), *youtube* (0.64%) and *news apps* (0.64%). Participants provided various triggers of their news encounter; from top to bottom were: *For relaxing or killing time* (35.17%), *To obtain more news information* (19.77%), *Out of habit* (19.63%), *For entertainment* (15.11%), *Notified by the phone* (6.92%), *For study or work purpose* (3.11%) and *For social* (0.28%).

Fig. 1 further shows triggers by news exposure. Participants' news exposure on news/content recommender system (e.g. Google

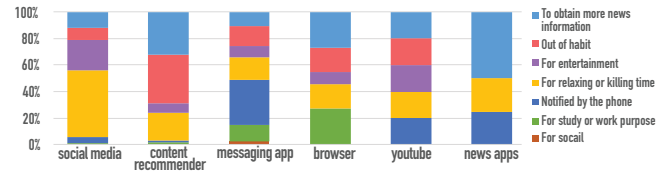


Figure 1: The percentage of using different apps to encounter news by various triggers

Discover) was more often triggered by users' habitual use of the phone or the motivation to know more about recent news; their news exposure on social media apps was more often triggered by the initial motivation for relaxing or entertainment; and finally, the triggers of their news exposure on messaging apps were highly diverse, but the top trigger was phone notifications. These results showed that news exposure on content recommender systems are more self-initiated, whereas other news exposure was more often serendipitous.

4 CONCLUSION AND FUTURE WORK

We have shown that news exposure on smartphones was mainly associated with different kinds of triggers. Our future work will continue to clarify the dynamic relationship among context, triggers, and news exposure. We will also investigate what follow-up actions users perform after different kinds of news exposures.

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